

MFE5200 – Computational Methods in Financial Engineering
Summer Term, 2025 – 26

Assignment 1

Due: 11:00 pm, 22-June-2026

Important notes:

1. The assignment must be submitted via Blackboard.
 2. The program codes of the programming problems must also be submitted.
 3. Total: 10 Problems (Full Mark: 100).
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Problem 1

Consider the following two Monte Carlo schemes for estimating

$$\mu = E[h(X) + f(X)].$$

The total sample size is $2n$ in both schemes.

Scheme 1: Use Same Random Numbers. Generate $2n$ i.i.d. copies of X , say $\{X_1, \dots, X_{2n}\}$. The estimator is

$$\hat{\mu}_1 = \frac{1}{2n} \sum_{i=1}^{2n} [h(X_i) + f(X_i)].$$

Scheme 2: Use Different Random Numbers. Write $\mu = E[h(X)] + E[f(X)]$ and estimate the two expected values separately. Generate $2n$ i.i.d. copies of X , say $\{X_1, \dots, X_n, Y_1, \dots, Y_n\}$. The estimator is

$$\hat{\mu}_2 = \frac{1}{n} \sum_{i=1}^n h(X_i) + \frac{1}{n} \sum_{i=1}^n f(Y_i).$$

Show that both $\hat{\mu}_1$ and $\hat{\mu}_2$ are unbiased estimators for μ , and

$$\text{var}(\hat{\mu}_1) - \text{var}(\hat{\mu}_2) = -\frac{1}{2n} \text{Var}[h(X) - f(X)].$$

Hence, deduce that $\text{var}(\hat{\mu}_1) \leq \text{var}(\hat{\mu}_2)$.

Problem 2

Given that $X(t)$ is a Geometric Brownian motion (GBM) which follows

$$\frac{dS(t)}{S(t)} = rdt + \sigma dW(t), \quad S(0) = S,$$

where $W(t)$ is a standard Brownian motion under the risk-neutral measure Q .

- (a) Let $Y(t) = S^\alpha(t)$. Find the stochastic differential equation of $Y(t)$.
(b) Use the SDE of $Y(t)$ in (a), suggest two Monte-Carlo methods to estimate

$$E^Q \left[e^{-rT} \max \left(\frac{\sum_{i=1}^m Y(t_i)}{m} - K, 0 \right) \right].$$

where $0 < t_1 < t_2 < \dots < t_m < T$.

One must be a biased estimator, and the other must be an unbiased estimator.

Problem 3

We simulate X from an inverse transform algorithm with the following way

$$X = \text{int}(10U(1 - U)),$$

where $\text{int}(x)$ = integral part of x , $U \sim \text{Uniform}(0, 1)$ and note that $0 \leq U \leq 1$.

Find $P(X=0)$, $P(X=1)$ and $P(X=2)$.

Problem 4

Suppose that

$$Y = (X - 3)^2 \quad \text{where} \quad X \sim \text{Binomial}(5, 0.5).$$

Find the probability mass function (pmf) of Y .

Problem 5

Let X be a random variable with the cdf of

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.1x, & 0 \leq x < 1 \\ 0.3, & 1 \leq x < 3 \\ 0.2x, & 3 \leq x \leq 5 \\ 1, & x > 5. \end{cases}$$

Determine the simulated values of x resulting from a uniform number u .

Problem 6

Suppose that we wish to generate samples from a probability distribution with density

$$f(x) = \begin{cases} \frac{1}{2}x^2e^{-x} & \text{if } x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

by the acceptance-rejection method. The alternative density function is chosen to be

$$g(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

for some $\lambda \in (0, 1)$. Determine the best λ that minimizes the expected number of trial samples needed to generate a sample from f .

Problem 7

Suppose that X has the standard double exponential distribution with the pdf of

$$g(x) = \frac{1}{2}e^{-|x|}, \quad -\infty < x < \infty.$$

Show that $|X|$ has the exponential distribution with mean 1.

Problem 8

Let Θ be uniform on $[0, 2\pi]$ and S is an exponential random variable with mean of 2. Assume that Θ and S are independent. Show that

$$X = \sqrt{S} \cos \Theta, \quad Y = \sqrt{S} \sin \Theta$$

are independent standard normal random variables.

Hint:

You may find the following result is useful:

Suppose that X_1 and X_2 are **continuous** random variables with joint probability distribution $f(x_1, x_2)$. Let $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ define a one-to-one transformation between the points (x_1, x_2) and (y_1, y_2) so that the equations $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ may be uniquely solved for x_1 and x_2 in terms of y_1 and y_2 , say $x_1 = w_1(y_1, y_2)$ and $x_2 = w_2(y_1, y_2)$. Then the joint probability distribution of Y_1 and Y_2 is

$$g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)]|J|,$$

where the Jacobian is the 2×2 determinant

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

and $\frac{\partial x_1}{\partial y_1}$ is simply the derivative of $x_1 = w_1(y_1, y_2)$ with respect to y_1 with y_2 held constant, referred to in calculus as the partial derivative of x_1 with respect to y_1 . The other partial derivatives are defined in a similar manner.

Problem 9

You are given that

$$\Sigma = \begin{pmatrix} 3.21 & 1.82 & 2.18 \\ 1.82 & 5.89 & 3.23 \\ 2.18 & 3.23 & 8.12 \end{pmatrix}.$$

Find the Cholesky factor A of Σ .

Problem 10

Let X and Y be two arbitrary random variables. The conditional variance of X given Y is defined as

$$\text{Var}[X|Y] = E[X^2|Y] - (E[X|Y])^2$$

Show that

- a. $\text{Var}[X|Y] \geq 0$.
- b. $\text{Var}[X] = E[\text{Var}[X|Y]] + \text{Var}[E[X|Y]]$.

Hint:

For any given functions $f(x)$, $g(x)$, $h(x)$ and $k(x)$, we have the following results

- i. $E[E[f(X)|Y]] = E[f(X)]$.
- ii. $E[g(E[h(X)|Y])k(X)|Y] = g(E[h(X)|Y]) E[k(X)|Y]$.

End